GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Computational Mathematics

DS-502-MJ

Sample / Practice Question Paper

Semester I · Core · Practice Paper

Prepared by Siddheshwar Thaware · Free for SPPU students

& This is a SAMPLE / PRACTICE question paper prepared for revision from the SPPU M.Sc. Data Science syllabus. It is NOT an official or past university examination paper. Use it only to test your preparation.

Time: 3 Hours Maximum Marks: 70

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

a) Define a vector space.
b) What is a basis of a vector space?
c) Define eigenvalue and eigenvector.
d) What is the rank of a matrix?
e) State the inner product of two vectors.
f) What is an orthogonal set?
g) Define a linear map.

Q2. Answer the following. [15]

A) Explain vector spaces, subspaces and linear independence with examples.

OR

B) Explain eigenvalues, eigenvectors and diagonalisation of a matrix.

Q3. Answer the following. [15]

A) Explain solving a system of linear equations using row reduction and echelon form.

OR

B) Explain Singular Value Decomposition (SVD) and QR decomposition.

Q4. Answer the following. [15]

A) Describe how matrix methods are applied in data science (SVD, NMF).

OR

B) Explain the Gram-Schmidt process and orthogonal projections.